File Integrity Monitoring System

A Cybersecurity Project using Python (Tkinter & SHA256 Hashing)

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Why This Project is Needed?

- This project is needed because in modern computer systems, data security is very important. Files can be accidentally changed or intentionally modified by hackers, viruses, or unauthorized users.
- **Problem:**

Most systems do not have a simple way to detect if a file has been changed or tampered with.

- **Risk:**
- Even a small change in an important file can lead to data loss, security breaches, or system corruption

Cybersecurity Concept Used:

Cryptographic Hashing (SHA256)

Technologies Used:

This project is developed using:

- **Python** (main programming language)
- **Tkinter** (for graphical user interface)
- **Hashlib** (for SHA256 hashing)
- **Threading** (for background file monitoring)
- **Time module** (for periodic checking)”

How It Works

- Select File
- Generate hash
- Store old Hash
- Compare new hash every few seconds
- If different --------- file changed

Weaknesses of This Project

- File must be manually selected
- No real-time system-wide monitoring
- Simple alert system
- No data storage
- Basic security level

Hash:

```
PS C:\Users\zoyac\Desktop\infosecproject> python -u "c:\Users\zoyac\Desktop\infosecproject\Usman.py"
Text: Usman Azeem
SHA256 Hash: 5dccc3d0d34253d1cd021ce0ef1a2be26124290094aee5dcde5f0e2348741bec
PS C:\Users\zoyac\Desktop\infosecproject>
```

Disadvantage of hashing?

Conclusion &Key takeaway:

- This project demonstrates a File Integrity Monitoring System using SHA256 hashing. It helps detect any unauthorized changes in files by comparing hash values.

- It ensures data integrity and improves basic file security by quickly identifying if a file has been modified